



Department of Computer Science
Faculty of Computing
UNIVERSITI TEKNOLOGI MALAYSIA

SUBJECT NAME :	COMPUTER ORGANIZATION AND ARCHITECTURE
SUBJECT CODE :	SECR 1033
SEMESTER :	2 - 2023/24
LAB TITLE :	Programming 1: Assembly Language Fundamentals
INSTRUCTION :	Student is required to have instructor's signature before proceed from each lab work.
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COMMENTS :	

Lab # 1

Simple program to familiarize with Program Code, Rebuild & Start Without Debugging

Execute the programs below:

PART 1:

i. Part A: Adding and Subtracting Integers

```
TITLE Add and Subtract (AddSub.asm)

; This program adds and subtracts 32-bit integers
; Authors:
; Date:
; Revision:

INCLUDE Irvine32.inc

.data
TOTAL dword 0 ; a variable named TOTAL (declared as DWORD)

.code

main PROC

    mov eax, 123400h ; Set EAX with the value of 123400h
    add eax, 567800h ; Add the content of EAX with 567800h
    sub eax, 77700h  ; Subtract content of EAX with 77700h
    mov TOTAL, eax  ; Store content of EAX to TOTAL

    call DumpRegs

    exit
main ENDP

END main
```

Experimental Results:

a) Rebuild & Start Without Debugging



Completed *Fully* *Partially* : Checked by: _____

Screenshot Result:

```
Microsoft Visual Studio Debug
EAX=00030000 EBX=006DC000 ECX=00E310AA EDX=00E310AA
ESI=00E310AA EDI=00E310AA EBP=008FFC4C ESP=008FFC40
EIP=00E3367B EFL=00000206 CF=0 SF=0 ZF=0 OF=0 AF=0 PF=1

C:\Users\Asus\source\repos\lab_template\lab 1 q2\Debug\lab 1 q2.exe (process 13772) exited with code 0.
Press any key to close this window . . .
```

ii. Part B: Adding Variables

```
TITLE Add and Subtract, Version 2 (AddSub2.asm)

; This program adds and subtracts 32-bit unsigned
; integers and stores the sum in a variable.
; Authors:
; Date:
; Revision:

INCLUDE Irvine32.inc
.data
val1 DWORD 10000h
val2 DWORD 40000h
val3 DWORD 20000h
finalVal DWORD ?
.code

main PROC
mov eax, val1      ; start with 10000h
add eax, val2      ; add 40000h
sub eax, val3      ; subtract 20000h
mov finalVal, eax ; store the result (30000h)

call DumpRegs     ; display the registers

exit

main ENDP
END main
```

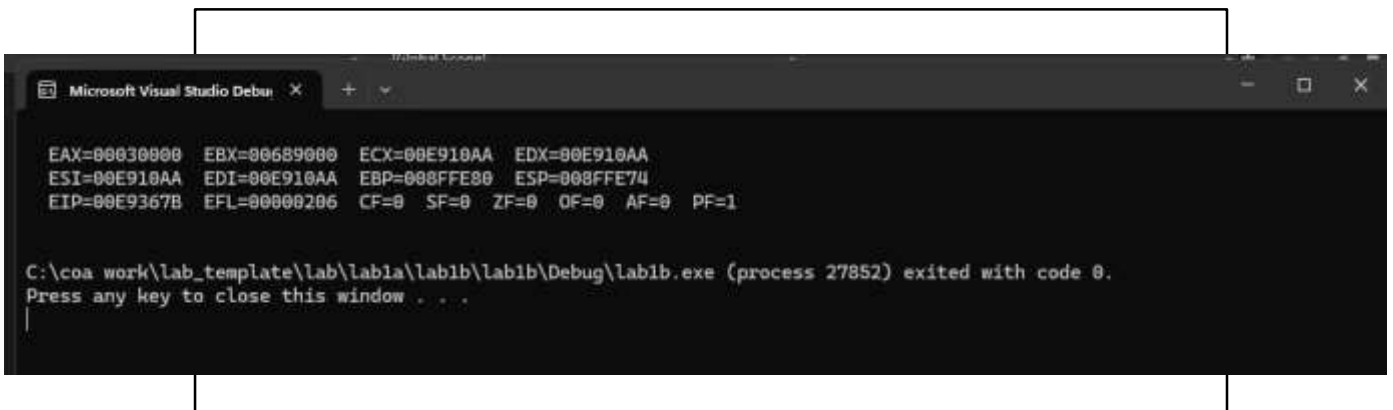
Experimental Results:

a) Rebuild & Start Without Debugging



Completed *Fully* *Partially* : Checked by: _____

Screenshot Result:



iii. Part C: Add and Subtract 8 and 16-Bit Version

```
TITLE Add and Subtract, Version 3
; This program adds and subtracts 8 and 16 bit
; unsigned integers and stores the sum in a variable.
; Authors:
; Date:
; Revision:

INCLUDE Irvine32.inc

.data
valw1 WORD 1000h
valw2 WORD 4000h
valw3 WORD 2000h
finalValw WORD ?

valb1 BYTE 10h
valb2 BYTE 40h
valb3 BYTE 20h
finalValb BYTE ?

.code
main PROC
mov ax,valw1; start with 10000h
add ax,valw2    ; add 40000h
sub ax,valw3    ; subtract 20000h
mov finalValw,ax ; store the result (30000h)
call DumpRegs  ; display the registers

mov ah,valb1; start with 10000h
add ah,valb2    ; add 40000h
sub ah,valb3    ; subtract 20000h
mov finalValb,ah ; store the result (30000h)
call DumpRegs  ; display the registers

exit
main ENDP
END main
```

Experimental Results:

a) Rebuild & Start Without Debugging



Completed *Fully* *Partially* : Checked by: _____

Screenshot Result:

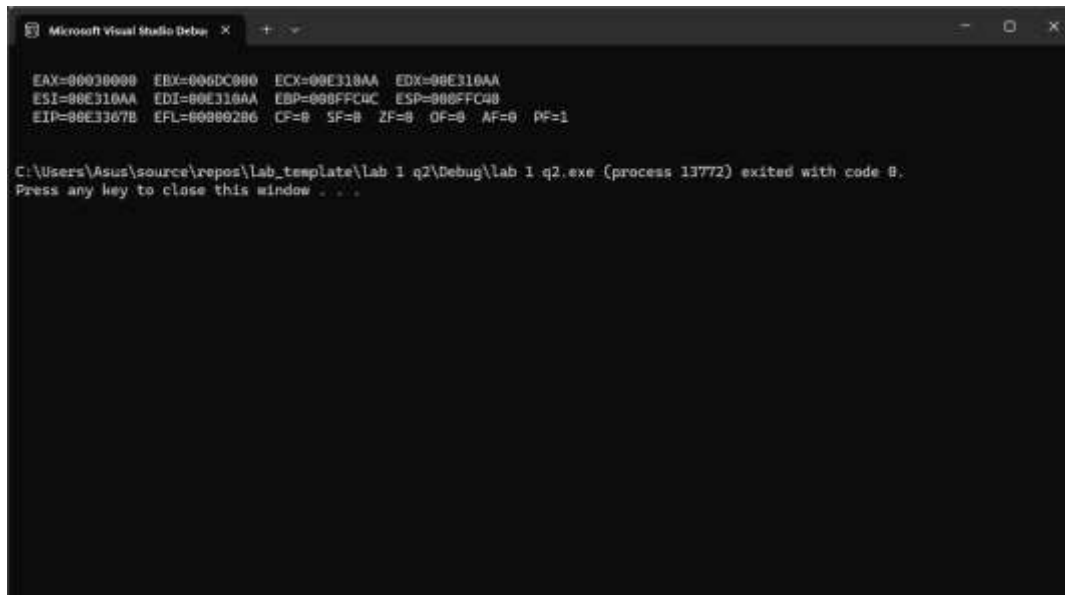
```
Microsoft Visual Studio Debug
EAX=00533800 EBX=006B3800 ECX=005C10AA EDX=005C10AA
ESI=005C10AA EDI=005C10AA EBP=0053FBE4 ESP=0053FB08
EIP=005C367F EFL=00000206 CF=0 SF=0 ZF=0 OF=0 AF=0 PF=1

EAX=00533800 EBX=006B3800 ECX=005C10AA EDX=005C10AA
ESI=005C10AA EDI=005C10AA EBP=0053FBE4 ESP=0053FB08
EIP=005C369C EFL=00000206 CF=0 SF=0 ZF=0 OF=0 AF=0 PF=1

C:\Users\Asus\source\repos\lab_1 (question c)\Debug\lab_1 (question c).exe (process 6768) exited with code 0.
Press any key to close this window . . .
```

ANSWERS AND SCREENSHOT

PART A:

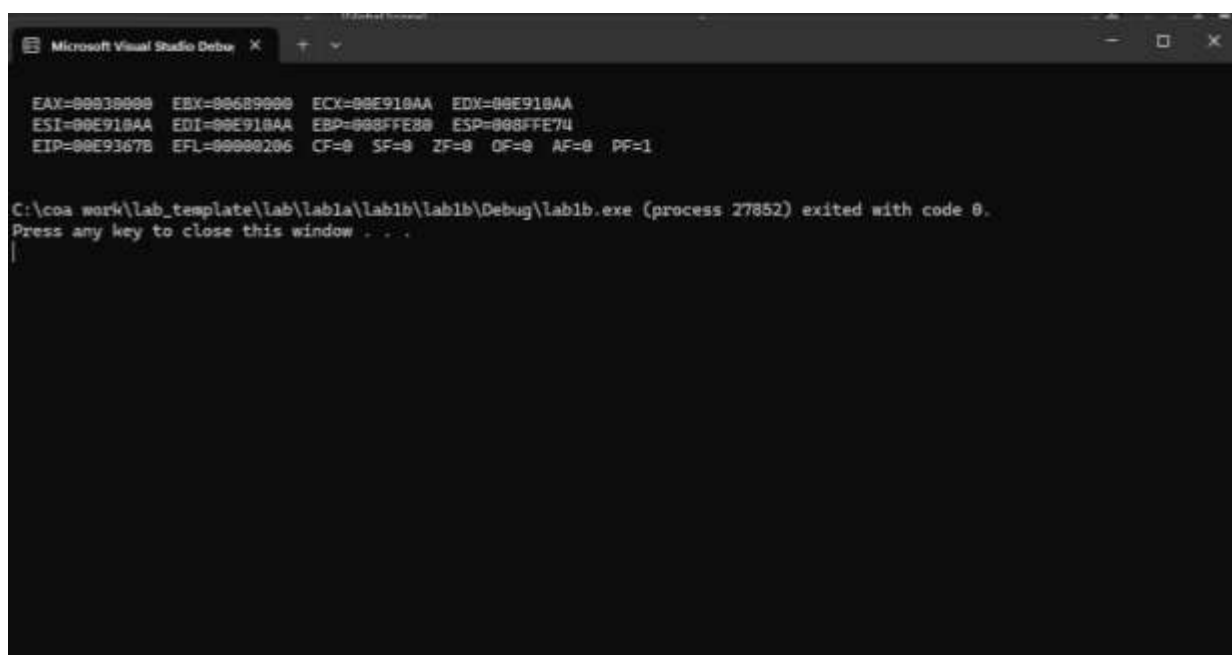


Microsoft Visual Studio Debug Console

EAX=00030000 EBX=006DC000 ECX=00E310AA EDX=00E310AA
ESI=00E310AA EDI=00E310AA EBP=000FFC4C ESP=000FFC40
EIP=00E3367B EFL=00000206 CF=0 SF=0 ZF=0 OF=0 AF=0 PF=1

C:\Users\Asus\source\repos\lab_template\lab 1 q2\Debug\lab 1 q2.exe (process 13772) exited with code 0.
Press any key to close this window . . .

PART B (WITHOUT DEBUGGING):

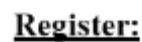


Microsoft Visual Studio Debug Console

EAX=00030000 EBX=00689000 ECX=00E910AA EDX=00E910AA
ESI=00E910AA EDI=00E910AA EBP=000FFE20 ESP=000FFE74
EIP=00E9367B EFL=00000206 CF=0 SF=0 ZF=0 OF=0 AF=0 PF=1

C:\cos work\lab_template\lab\lab1a\lab1b\lab1b\Debug\lab1b.exe (process 27852) exited with code 0.
Press any key to close this window . . .

Memoryy



PART C (WITHOUT DEBUGGING):

```
Microsoft Visual Studio Debu
+ - + - +

EAX=00533000 EBX=004E3000 ECX=005C10AA EDX=005C10AA
ESI=005C10AA EDI=005C10AA EBP=0053FB08 ESP=0053FB08
EIP=005C107F EFL=00000206 CF=0  SF=0  ZF=0  OF=0  AF=0  PF=1

EAX=00533000 EBX=004E3000 ECX=005C10AA EDX=005C10AA
ESI=005C10AA EDI=005C10AA EBP=0053FB08 ESP=0053FB08
EIP=005C109C EFL=00000206 CF=0  SF=0  ZF=0  OF=0  AF=0  PF=1

C:\Users\Asus\source\repos\lab_1 (question c)\Debug\lab_1 (question c).exe (process 6768) exited with code 0.
Press any key to close this window . . .
```

The screenshot shows the Visual Studio IDE with the following components:

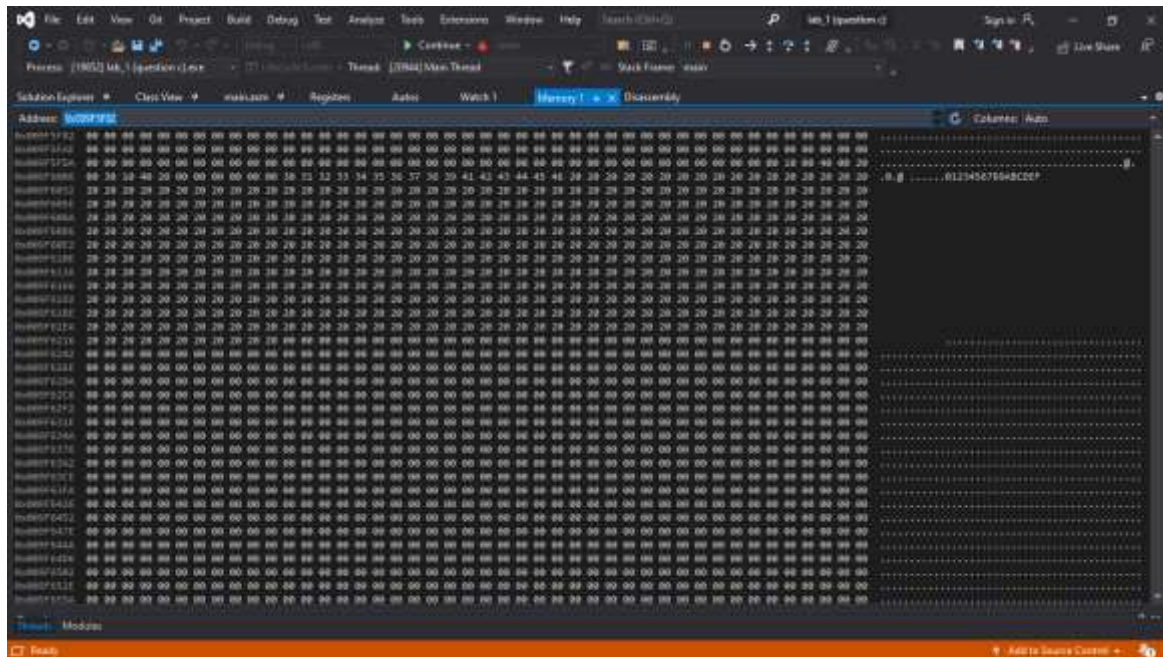
- Menu Bar:** File, Edit, View, Go, Project, Build, Debug, Tools, Extensions, Window, Help.
- Toolbar:** Standard Visual Studio toolbar with icons for file operations, editing, and debugging.
- Solution Explorer:** Located on the left, it shows the project structure. The 'Lab 1' project is selected, and the 'Assembly' file is highlighted.
- Code Window:** The main area on the right displays the assembly code for 'Lab 1'. The code is written in x86 assembly and includes comments in English. The code is as follows:


```

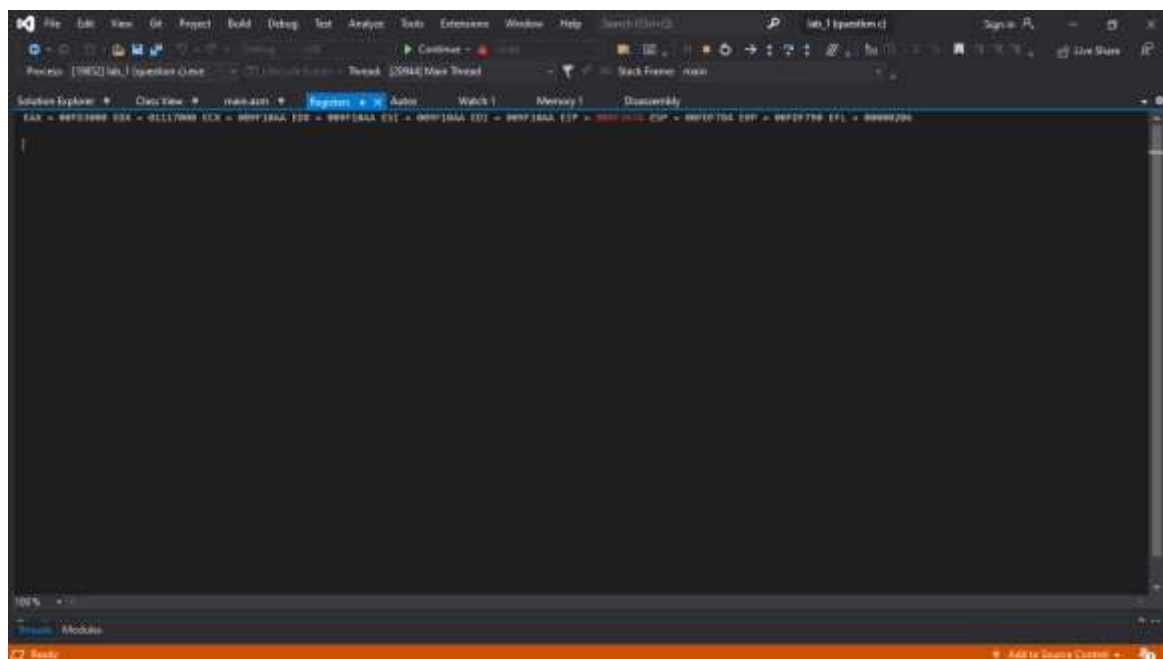
1  TITLE Added Subtract, Version 1
2  ; This program performs subtracts 8 and 16 to
3  ; assigned integer and stores the result in a variable.
4  ; Authors: adrian and alish
5  ; Date: 28 April 2018
6  ; Section: Lab 1
7
8  INCLUDE Irvine32.inc
9
10 .code
11
12 value WORD 1000h
13 value WORD 4000h
14 value WORD 2000h
15 finalValue WORD ?
16 value BYTE 10h
17 value BYTE 40h
18 value BYTE 30h
19 finalValue BYTE ?
20
21 .start
22 main PROC
23     mov ax, value; start with 1000h
24     sub ax, value; sub 4000h
25     mov ax, value; subtract 2000h
26     mov finalValue, ax; store the result(1000h)
27     call DumpRegs; display the registers
28
29     mov ax, value; start with 1000h
30     sub ax, value; sub 4000h
31     mov ax, value; subtract 2000h
32     mov finalValue, ax; store the result(1000h)
33     call DumpRegs; display the registers
34
35     exit
36 main ENDP
37
38 END main
      
```
- Status Bar:** Located at the bottom, it shows the current line (10), column (10), and other information.

PART C (DEBUGGING):

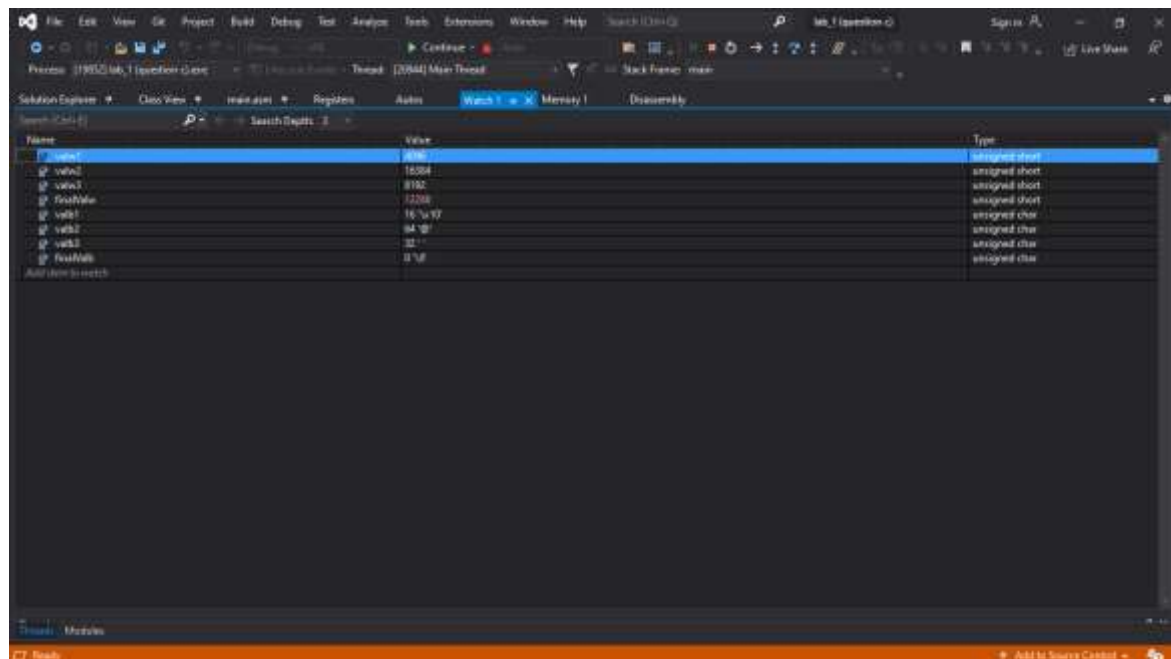
Memory



Register:



Watch:



Autos:

